

Population and Sampling Methods

Introduction

In Statistics, we often want to study a large group of people or objects. Studying every individual may be difficult, expensive, or time-consuming. Therefore, we often select a smaller group from the larger group.

Population

A **population** is the complete group of people, objects, or observations that we want to study.

Example: All Zomato users in Ahmedabad.

Sample

A **sample** is a smaller group selected from the population for collecting information and making conclusions about the entire population.

Example: 100 randomly selected Zomato users from Ahmedabad.

Q1. Real-World Examples of Population and Sample

A. Examples Where I Am Part of a Population

Example 1: Instagram

Suppose Instagram wants to study the average time spent on Instagram by users in India.

Population: All Instagram users in India.

I am one of the users in this complete group, so **I am part of the population**.

Example 2: Zomato

Suppose Zomato wants to study customer satisfaction among users in Ahmedabad.

Population: All Zomato customers in Ahmedabad.

If I use Zomato, I am included in this population.

Example 3: Flipkart

Suppose Flipkart wants to study the shopping preferences of its customers.

Population: All Flipkart customers.

If I am a Flipkart customer, I am one member of this population.

B. Examples Where I Am Part of a Sample

Example 1: Instagram Survey

Instagram wants to survey **1,000 users** about a new feature.

If I am randomly selected among those 1,000 users:

Sample: 1,000 selected Instagram users.

I am therefore a member of the sample.

Example 2: Zomato Restaurant Survey

Zomato selects **10 restaurants** from Ahmedabad to study restaurant ratings.

If I am a customer whose feedback is selected for the survey, I am part of the sample.

Sample: Selected customers providing feedback.

Example 3: Flipkart Customer Survey

Flipkart selects **500 customers** to ask about their shopping experience.

If I am one of the selected 500 customers:

Sample: 500 selected Flipkart customers.

Therefore, I am part of the sample.

Summary Table

Situation	Population	Sample
Instagram	All Instagram users in India	1,000 selected users
Zomato	All Zomato customers in Ahmedabad	Selected customers
Flipkart	All Flipkart customers	500 selected customers

Key Difference

Population = Entire group

Sample = Smaller group selected from the population

Q2. Random Sampling Plan for Zomato

Problem

Zomato wants to find the **average rating of restaurants in Ahmedabad**.

It wants to select **10 restaurants** for a survey.

Random Sampling Plan

Step 1: Prepare the Population

First, Zomato should create a complete list of all restaurants in Ahmedabad that are included in the study.

For example:

$R_1, R_2, R_3, \dots, R_{1000}$

Suppose there are 1,000 restaurants.

Step 2: Assign Numbers

Give each restaurant a unique number:

Restaurant 1
Restaurant 2
Restaurant 3
...
Restaurant 1000

Step 3: Randomly Select 10 Restaurants

Use a random number generator or lottery method to select **10 different numbers** from 1 to 1,000.

For example, the selected numbers might be:

25, 87, 143, 219, 305, 421, 578, 650, 812, 934
25, \ 87, \ 143, \ 219, \ 305, \ 421, \ 578, \ 650, \ 812, \ 934

The corresponding restaurants become the sample.

Step 4: Collect Ratings

Zomato can collect ratings or survey responses from these 10 selected restaurants/customers, depending on the study design.

Why Is Random Sampling Important?

Random sampling gives restaurants a more equal opportunity to be selected. It reduces the chance of **selection bias**, where the researcher might unintentionally choose only popular, nearby, or highly rated restaurants.

Therefore, random sampling helps produce a sample that is **more representative of the population**.

Conclusion

A random sample of 10 restaurants can provide useful information about the larger restaurant population, although a sample of only 10 may still be small for estimating all of Ahmedabad accurately.

Q3. Flipkart Survey and Sampling Bias

Situation

Flipkart wants to survey users about a new feature.

However, the survey is sent **only to users who made a purchase last week**.

Type of Sampling

This is a form of **convenience sampling / restricted sampling**, because Flipkart is selecting users who are easily available within a particular group rather than giving every user an equal chance of selection.

The selection is restricted to:

Users who made a purchase last week.

Therefore, users who did not purchase anything last week have no chance of being selected.

Possible Source of Bias

This approach can create **selection bias**.

Users who made a purchase recently may have different opinions from users who have not purchased recently.

For example:

- Recent buyers may use shopping features more frequently.
- They may have more recent experience with Flipkart.
- Inactive or occasional users are completely excluded.

Therefore, the survey may not accurately represent the opinions of **all Flipkart users**.

Conclusion

The sample is not fully representative because **not every Flipkart user has an equal opportunity to participate**. This can lead to selection bias and may produce misleading conclusions about the new feature.

Q4. Customer Feedback Survey

Selected App: Instagram

Suppose Instagram has introduced a new feature and wants to understand whether users are satisfied with it.

The goal is to collect feedback from **20 users**.

Objective of the Survey

To understand user satisfaction and experience with the new Instagram feature.

Sampling Method: Stratified Sampling

I would use **stratified sampling** because Instagram has different types of users with different levels of activity.

For example, users can be divided into three groups:

1. **Low-activity users**
2. **Medium-activity users**
3. **High-activity users**

Instead of selecting all 20 users from only one group, Instagram can select users from each group.

Example

Suppose the sample of 20 users is divided as:

User Group	Number Selected
Low-activity users	6
Medium-activity users	8
High-activity users	6
Total	20

Users within each group can then be selected randomly.

Why Stratified Sampling?

Different types of users may have different experiences with Instagram.

For example:

- Low-activity users may find the feature difficult to discover.

- Medium-activity users may use the feature regularly.
- High-activity users may have more detailed opinions because they use Instagram frequently.

If we select users from all three groups, we can collect a **more balanced representation of user experiences**.

Therefore, stratified sampling is appropriate because it ensures that **different user groups are represented in the sample**.

Customer Feedback Survey Questions

The selected 20 users can be asked the following questions.

Q1. How often do you use the new feature?

- Daily
- Several times a week
- Occasionally
- Rarely

Q2. How easy is the feature to use?

- Very Easy
- Easy
- Neutral
- Difficult
- Very Difficult

Q3. How satisfied are you with the feature?

Rate from:

1=Very Dissatisfied1=\text{Very Dissatisfied}

to

5=Very Satisfied5=\text{Very Satisfied}

Q4. Does the new feature improve your Instagram experience?

- Yes
- No
- Not Sure

Q5. What improvement would you suggest?

Open-ended response

Sampling Plan

The complete process would be:

All Instagram Users



Divide users into groups



Low / Medium / High Activity



Randomly select users
from each group



20 Selected Users



Conduct Feedback Survey



Analyze Responses